

Vacuum Cleaner Problem Using Python

Aim

To write a Python program to simulate the **Vacuum Cleaner Problem**, where a vacuum cleaner cleans all dirty rooms by moving between rooms until every room is clean.

Algorithm

1. Start the program.
2. Initialize the rooms with their current status (Dirty/Clean).
3. Place the vacuum cleaner in the initial room.
4. Check whether the current room is dirty.
5. If the room is dirty, clean it.
6. Otherwise, move the vacuum cleaner to the other room.
7. Repeat Steps 4–6 until all rooms are clean.
8. Display the message "All rooms are clean!".
9. Stop the program.

Python Code

```
def vacuum_cleaner():
    rooms = {'A': 'Dirty', 'B': 'Dirty'}
    location = 'A'
    while 'Dirty' in rooms.values():
        print("Location:", location, "|",
              Status:", rooms[location])
        if rooms[location] == 'Dirty':
            rooms[location] = 'Clean'
            print("Cleaning Room",
                  location)
        else:
            location = 'B' if location ==
            'A' else 'A'
            print("Moving to Room",
                  location)
        print("\nAll rooms are clean!")
    vacuum_cleaner()
```

EXP 6(Vacuum Cleaner Problem).py > ...

```
1 def vacuum_cleaner():
2     rooms = {'A': 'Dirty', 'B': 'Dirty'}
3     location = 'A'
4     while 'Dirty' in rooms.values():
5         print("Location:", location, "| Status:", rooms[location])
6         if rooms[location] == 'Dirty':
7             rooms[location] = 'Clean'
8             print("Cleaning Room", location)
9         else:
10            location = 'B' if location == 'A' else 'A'
11            print("Moving to Room", location)
12    print("\nAll rooms are clean!")
13 vacuum_cleaner()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\OPEN CV\AI Lab> & C:\Users\user\AppData\Local\Programs\Python\Python310\pyth
Location: A | Status: Dirty
Cleaning Room A
Location: A | Status: Clean
Moving to Room B
Location: B | Status: Dirty
Cleaning Room B

All rooms are clean!
```